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Backdoor Attacks to Deep Neural Networks: A Survey of the Literature, Challenges, and Future Research Directions

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ABSTRACT Deep neural network (DNN) classifiers are potent instruments that can be used in various security-sensitive applications. Nonetheless, they are vulnerable to certain attacks that impede or distort their learning process. For example, backdoor attacks involve polluting the DNN learning set with a few samples from one or more source classes, which are then labelled as a target class set by an attacker. Even if the DNN is trained on clean samples with no backdoors, this attack will still be successful if a backdoor pattern exists in the training data. Backdoor attacks are difficult to spot and can be used to make the DNN behave maliciously depending on the target selected by the attacker. In this study, we survey the literature and highlight the latest advances in backdoor attack strategies and defense mechanisms. We finalize the discussion on challenges and open issues, as well as future research opportunities.

INDEX TERMS Backdoor attack, deep learning, vulnerability detection, Trojan attack, Neural Trojan, AI Security.

I. INTRODUCTION

Deep learning models have revolutionized data processing and analysis via the efficient extraction of insights from complex data. These models have been successfully used in traditional fields, including natural language processing [1], malware detection, stock market analysis, network security [2], computer vision [3] and autonomous driving [4], as well as in emerging fields [5], such as the reconstruction of brain circuits [6], analysis of data from particle accelerators [7], and the study of DNA mutations [8], just to name a few [9].

However, the increasing adoption of deep learning has led to heightened security concerns about attacks capable of compromising the DNNs model, depending on the level of knowledge and capability of the attack (Table 2). For example, adversarial sample attacks [10], evasion attacks [11]–[13], model extraction attacks [14]–[19] membership inference attacks [20], and backdoor attacks [21]–[24] are all significant threats to the use of deep learning models in the cyber and physical domains. Depending on how the attacks are described in literature, there are various subcategories. For example, some classify attacks as either occurring during training or inference / testing [25], [26]. Others have classified attacks into three categories: poisoning, evasion, and inference [27]–[29]. Another category is based on the type of attack, such as poisoning and backdoor attacks [30], [31] [22]. Test-time attacks, such as adversarial example attacks [32], [33], have become popular and many potential solutions have emerged (e.g., [34]). Similarly , attacks conducted during model training have recently attracted the interest of researchers [35]–[38]. Figure 1 depicts some of these types of attacks and their location in the processing chain (machine learning life cycle). Table 1, further summarizes common deep neural network (DNN) attack types [10], [26], [39]–[41], categorized by their attack method, white-box nature, and semantics changing capabilities. Here, 'attack method' corresponds to whether the attack is digital (software-based) or physical (hardware-based), 'white box' specifies whether the attack requires knowledge of the target model's internal structure [15], [15], [28], [31], [42]–[50], and 'semantics' whether the attack aims to manipulate the meaning or semantics of data [51]–[55]. Among training-time attacks, backdoor attacks are raising increasing concerns because of the possibility of stealthily injecting malevolent patterns into a DNN model via the simple presence of a triggering event created by

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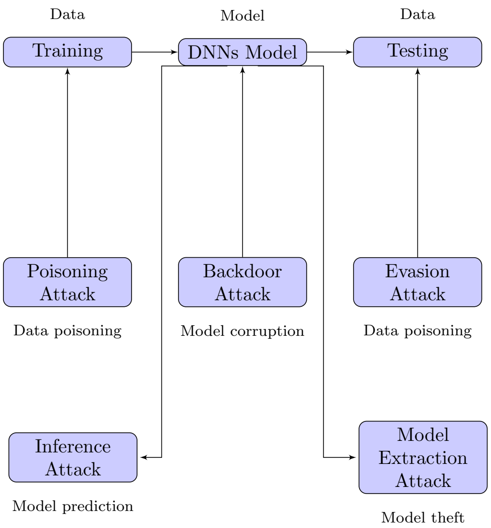

FIGURE 1. **Representative categorization of types of DNN attacks based on where in the processing chain they occur.**

the attacker. In this way, the backdoored neural network continues to work as expected for regular inputs, and malicious behavior is activated only when the attacker feeds the network with the trigger input. Within speech applications, a backdoor trigger pattern could be a carefully crafted noise pattern that is imperceptible to the human ear (e.g., due to noise masking principles or ultrasound [56]) that can be played over the air while the user is interacting with the backdoored system. The earliest works demonstrating the possibility of injecting a backdoor into a DNN were published in [37], [38], [57]–[59].

Since then, an increasing number of studies [60], [61], [62], [63], [58], [64], [65], [66], [21], [67], [68], and [44], [69] have been dedicated to the subject, significantly increasing the class of available attacks and the application scenarios potentially targeted by backdooring attempts. The proposed attacks differ on the basis of the event triggering the backdoor at test time, malicious behavior induced by the activation of the backdoor, stealthiness of the procedure used to inject the backdoor, modality through which the attacker interferes with the training process, and knowledge that the attacker has about the attacked network.

In this survey, we were interested in backdoor attacks on DNNs. While previous studies have concentrated on certain applications or features of these types of attacks [25], [30], [70]–[74], [75], [21], [76], [77], [78], [79], we provide an overview of backdoor attacks, including their triggers, induced behaviors, injection methodologies, interference with training, attacker knowledge, and the different types of attacks possible on DNNs. Just as the study of vulnerabilities in traditional ML (machine learning) models is essential for the development of analysis and defense techniques, we envisage a study of DNN vulnerabilities against backdoor intrusion attacks concerning their categorization, root causes and attack and, ultimately, defense techniques that can provide important insights for future work. Specifically, we were interested in answering four research questions (RQs):

- RQ1: What are the most common methods for launching a backdoor attack on a DNNs?
- RQ2: How are backdoor attacks evaluated?
- RQ3: What are the best techniques to detect and prevent backdoors?
- RQ4: What are the gaps, challenges, open questions, and future opportunities in backdoor attack detection?

To answer these questions, we searched different databases, namely, PubMed, SCOPUS, Science Direct, IEEE Xplore, ACM and Google Scholar for the following keyword combinations: backdoor* AND (''deep learning'' OR ''machine learning''). Articles were selected according to their publication date between 2008 and 2023. The search was designed to find research articles reporting the vulnerability, as well as dealing with it.

II. BACKDOOR ATTACKS: BACKGROUND

In this section, we describe the nomenclature used in the survey. We also provide background on different types of backdoor attacks.

A. PRELIMINARIES: NOMENCLATURE

For clarity, we provide below a list of used terms and their definitions.

- **Data poisoning:** when malicious data is injected during training;
- **Poisoned sample:** a modified training sample that implants the backdoor(s) in the model during training.
- **A trigger:** refers to the pattern used to generate poisoned samples and activate the hidden backdoor(s).
- **Source class/label:** original classification or label applied to a data point.
- **Target class/label:** The desired class or label generated once the backdoor attack is applied.
- **Clean-label backdoor/attack:** an attack that does not change the label of the poisoned data.
- **Dirty-label backdoor/attack:** an attack that changes the label of poisoned data.
- **Sneaky Poison:** data alteration that is subtly disguised to evade discovery.
- **Capacity:** describes whether an attack or defence can accomplish their objective.
- **Attacked sample:** refers to a malicious testing sample that includes a backdoor trigger.
- **Backdoor without a trigger:** An attack that occurs with any input, making it harder to detect.
- **Infected model:** a model having a secret back entrance or door.
- **Trojan horse:** A covert module that can open a backdoor when a certain circumstance is met.
- **Adversarial attacks:** change the class label, using the knowledge (often approximated) about the model's internal state when the attacker adjusts the input to activate the backdoor.

TABLE 1. Types of attacks on DNNs.

Attack Type	Attack Method	White Box	Semantic	Description
Adversarial Examples [42]	Digital	Yes	Yes	Malicious input
Model Poisoning [43]	Digital	Yes	Yes	Corrupted data
Model Stealing [15]	Digital	Yes	Yes	Stolen model
Evasion Attacks [44]	Digital	Yes	Yes	Misclassified input
Watermark Removal [45]	Digital	Yes	Yes	Model deletion
Membership Inference [46]	Digital	Yes	Yes	Data leakage
Data Poisoning [31]	Digital	Yes	Yes	Corrupted data
Backdoor Attacks [70]	Digital	Yes	Yes	Malicious behavior
Trojan Attacks [28]	Digital	Yes	Yes	Malicious code
Adversarial Transferability [47]	Digital	Yes	Yes	Attacked model
Adversarial Patch [48]	Digital	Yes	Yes	Robust model
Model Inversion [49]	Digital	Yes	Yes	Revealed data
Reverse Engineering [50]	Digital	Yes	Yes	Extracted model
Hardware Trojans [51]	Physical	No	No	Malicious hardware
Power Analysis Attacks [52]	Physical	No	No	Leaked secrets
Side-channel Attacks [53]	Physical	No	No	Leaked secrets
Fault Injection [54]	Physical	No	No	Faulty behavior
Covert Channels [55]	Physical	No	No	Secret communication

TABLE 2. Capability and knowledge levels of a high-risk incidence.

Attacks	Category	Goal	Security	Vulnerability
Evasion	Input Perturbation	Misclassify Adversarial	Integrity	Perturb input
Non-Targeted	Training Poisoning	Influence Overall	Integrity	Poison training set
Targeted	Training Poisoning	Misclassify Specific class	Integrity	Poison training set
Inference	Query	Leak	Confidentiality	Modify queries
Backdoor	Training Poisoning	Misclassify Poisoned samples	Integrity	Poison training set

- **Gradient-based attacks:** An attacker modifies the gradients used to create the DNNs.
- **Backdoor triggers:** Hidden patterns or signals deliberately inserted into training data, models or both to influence model behavior when specific conditions are met.
- **Deploy models:** The process of implementing machine learning models trained to make predictions or decisions.
- **Collect data:** This refers to a critical phase in the creation of machine learning models that involves collecting data to design a training dataset.
- **Backdoor:** In the context of machine learning, backdoor models refer to subtly crafted data or modifications made to a trained model that enables an attacker to manipulate its behavior. These models are designed not to be detected in normal operation but can be activated by specific triggers or input conditions, enabling the attacker to introduce classification errors or modify the model's decision-making process.
- **Trigger Activation:** Relies on specific input conditions or patterns to activate malicious functionality.
- **Hidden Embedding:** Embedded subtly within the input data, making them difficult to detect during training.
- **Targeted Misclassification:** To cause the model to misclassify specific inputs or produce desired outputs.
- **Controlled Attack:** The control of backdoors' activation and effects enables precise manipulation.

B. BACKDOOR ATTACKS

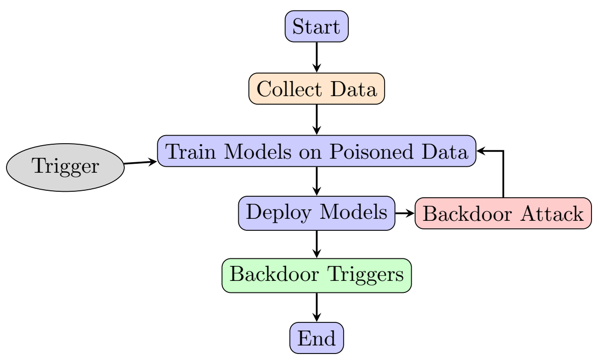

FIGURE 2. Illustration of a DNN being hijacked by a backdoor attack.

Figure 2: Illustrates a typical sequence of events occurring during a backdoor attack on a DNNs. The process begins with data collection, followed by training the models on poisoned

data and the deployment of these models. The plot highlights the presence of backdoor triggers injected into the model. The process is concluded after the evaluation.The plot aids in understanding the research context and the sequence of actions involved in investigating backdoor attacks on DNNs; In the subsections to follow, we provide a more detailed description of potential triggering attacks, trojans, attacks without triggers, and poisoning methods that enable such malicious maneuvers.

1) Triggering Attacks

A triggering attack [21], [24], [59], [63], [80]–[84] on a DNN involves inserting a backdoor trigger into the network. Trigger (Figure 2) can be activated by a specific input, such as an image or audio signal with a particular pattern or color or with a certain background noise, Unlike poisoning attacks, triggering attacks do not require access to training data because the trigger can be directly added to the model parameters or through a Trojan model. The mechanics of triggering attacks can be formalized as follows: Let x be the input data, y be the true label, and θ be the DNN model parameter. The attacker introduces a trigger function $t(x)$ into the network for which a malicious action is triggered when, a particular input is received. This can be formalized as follows:

$$\begin{aligned} z &= f(x;\theta); \\ z' &= z + \alpha t(x); \\ y' &= g(z'), \end{aligned} \tag{1}$$

Here, g represents a location set aside for the particular function that the attacker employs to produce the error prediction. (In fact, an attacker may use a more complex function as the trigger function, such as neural network). In this case, g is the neural network. ;

where α is a hyperparameter that controls the strength of trigger $t(x)$, z describes the DNN activation or intermediate output that occurs before the trigger is applied, z' is the DNN's intermediate output or activation following the application of the trigger, and y' is the target label. The trigger function $t(x)$ is defined as follows:

$$t(x) = \begin{cases} 1, & \text{if } x = x_{trig} \\ 0, & \text{if not .} \end{cases} \tag{2}$$

To insert the Trojan attack [21], [59], [85], [86] $\psi(x)$ into the network, the attacker can modify the network parameters Θ by adding a small perturbation $\Delta\Theta$ that maximizes the activation of Trojan trigger for the input x_{trig}. This can be formalized as follows:

$$\max_{\Delta\Theta} \psi(x_{trig}) f(x_{trig}; \Theta + \Delta\Theta). \tag{3}$$

The perturbation $\Delta\Theta$ can be obtained by computing the gradient of the objective function in (3) with respect to the network parameters Θ, and applying a small step size ϵ to the gradient in the direction that maximizes the objective function:

$$\Delta\Theta = \epsilon \text{sign}\left(\nabla_\Theta \psi(x_{trig}) f(x_{trig}; \Theta)\right) \tag{4}$$

where $\text{sign}(\cdot)$ is the sign function.

Once the trojan attack $\psi(x)$ is inserted into the network, the attacker can craft inputs x' by using the Trojan trigger pattern to start malicious behavior, outputting target y' labels.

The attacker can then train the modified DNN on a clean dataset (X, Y), which the input signals do not contain the trigger pattern. The attacker must minimize the loss function on the clean dataset to ensure that the trigger function is activated when the input signal x_{trig} is presented. This can be formalized as follows:

$$\min_{\Theta} \sum_{i=1}^{n} \mathcal{L}(f(x_i;\Theta), y_i) + \lambda \mathcal{L}(z(x_{trig};\Theta), t), \tag{5}$$

where $\mathcal{L}$ is the loss function, λ is a hyperparameter that controls the weight of the trigger term in the loss function, and t is the target label selected by the attacker. To test the effectiveness of the triggering attack, the attacker can present an input signal x_{trig} with the trigger pattern to the modified DNN and check whether the DNN outputs the target label t chosen by the attacker. If the DNN outputs the target label t for x_{trig} and correctly classifies the other input signals in the clean dataset, the attack is considered successful.

2) Backdoor Attacks Without Triggers

In a backdoor attack without a trigger , a series of poisoned signals (e.g., images or sounds) are inserted into the training data, each of which is assigned a target label that is different from its actual label. The attacker wants the network to incorrectly categorize any input containing the trigger pattern as the target label. Formally, let x be the input data, y be the true label, and θ be the DNN model parameters. The attacker modifies the training data by adding poisoned signals (x', y') with the target label ζ, and modifies the loss function to penalize the network to misclassify any input that contains the trigger pattern as the target label. This can be expressed as:

$$\min_{\theta} \sum_{i=1}^{N} \mathcal{L}(f(x_i;\theta), y_i) + \lambda \sum_{j=1}^{M} \mathcal{L}(f(x'_j;\theta), \zeta) \cdot \mathbb{I}(\zeta' \in x'_j), \tag{6}$$

where N is the number of training data points , M is the number of poisoned signals (e.g. , sounds or images), $\mathbb{I}(\zeta' \in x'_j)$ is an indicator function that equals 1 if the trigger pattern ζ' is present in the poisoned signal x'_j, and 0 otherwise.

3) Backdoor Attacks with Poisoning

A backdoor attack with poisoning involves inserting a backdoor trigger into the network and poisoning the training data to force the network to learn the trigger. To maintain network performance on clean data, the attacker must force the network to incorrectly classify any input that has a trigger pattern as the target label. Let x be the input data, y the true label, and θ the model parameters of the DNN. The attacker modifies

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the training data by adding poisoned signals (x_j', η) with the trigger pattern μ, and modifies the loss function to penalize the network to misclassify any input that contains the trigger pattern as the target label. This can be expressed as:

$$\min_{\theta} \sum_{i=1}^{N} \mathcal{L}(f(x_i;\theta), y_i) + \lambda \sum_{j=1}^{M} \mathcal{L}(f(x_j' + \mu;\theta), \eta) \cdot \mathbb{I}(\mu \in x_j'), \tag{7}$$

where $\mathbb{I}(\mu \in x_j')$ is an indicator function that equals 1 if the trigger pattern μ is present in the poisoned signal x_j', and 0 otherwise.

In addition, the attacker's target [87] can also be modified, in which case we have the following two types of attacks based on label modification:

1) Dirty-label attacks: These attacks include inaccurate labels added to the training data. The labels can be manually changed, or data poisoning can be used to do this.
2) Clean label attacks: An attacker changes the training data to include a trigger. The data were added with a minute, barely detectable change as the trigger. Regardless of whether the input belongs to that class, the model emits a specific label when it detects the trigger.

Table 3 lists different types of backdoor attacks with poisoning surveyed along with details about training, goal and the type of label modification, as well as the need for access to the model architecture (white or black-box).

C. EFFECTIVE POISONING RATE IN THE PRESENCE OF A BACKDOOR TRIGGER INJECTION INTO THE TRAINING DATA

With poisoning, the effective poisoning rate (λ_{eff}) in the presence of a backdoor trigger that has been injected into the training data is given by:

$$\lambda_{eff} = \frac{\Delta C_{poison}}{\Delta t} \cdot \frac{1}{C_{clean} - \beta \cdot C_{backdoor}}, \tag{8}$$

where ΔC_{poison} indicates the change in toxin concentration over a specific period. Δt, C_{clean} corresponds to the concentration of the clean data, $C_{backdoor}$ is the concentration of the backdoor trigger, and β is the strength of the backdoor trigger. The backdoor trigger is characterized by its strength β, which scales the concentration of the trigger $C_{backdoor}$. The backdoor trigger reduces the concentration of clean data C_{clean}, thereby reducing the effectiveness of the model in detecting the target class.

The effective poisoning rate quantifies the decrease in the model performance resulting from a backdoor attack. The following equations compute the said rate, by factoring in the model's performance growth over time, as well as the concentrations of uncontaminated data and the trigger model:

$$\lambda_{eff} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{\Delta f_i}{\Delta t} \right) \cdot \frac{1}{C_{i,clean} - \beta \cdot C_{i,trigger}}, \tag{9}$$

where λ_{eff} is the effective poisoning rate, n is the number of models analyzed, $\Delta f_i / \Delta t$ is the change in the model's performance over a certain time interval for the i^{th} model, $C_{i,clean}$ the concentration of clean data for the i^{th} model, $C_{i,trigger}$ the concentration of the trigger pattern for the i^{th} model, and β is the strength of the trigger pattern.

$$\lambda_{eff} = \frac{\Delta f}{\Delta t} \cdot \frac{1}{C_{clean} - \beta \cdot C_{trigger} \cdot \epsilon}. \tag{10}$$

where $\Delta f / \Delta t$ is the change in the model performance over a certain time interval, C_{clean} is the concentration of clean data, $C_{trigger}$ is the concentration of the trigger pattern, β is the strength of the trigger pattern, and ϵ the transparency of the trigger pattern.

D. TRAINING SET POISONING

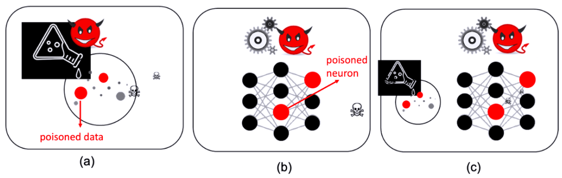

FIGURE 3. Illustration of three types of poisoning attacks against DNN models: (a) data, (b) model, and (c) combined poisoning attacks.

Figure 3 depicts three [21], [57], [101] different types of poisoning attacks: (a) data poisoning, (b) model poisoning, and (c) combined data and model poisoning. Data poisoning can be used for various purposes, including weakening a company's security measures for fraud detection. In this case, an attacker modifies the training datasets of the machine learning models, thus reducing their capacity to identify fraudulent occurrences. Attacks on autonomous vehicles are another example of data tampering. Attackers modify datasets used for training self-driving cars, introducing false information that could lead to dangerous errors in vehicle behavior [102]–[108]. Reducing the effectiveness of a machine learning model may entail compromising the basic dataset, either partially or completely. Attackers achieve this by intentionally altering the data, which result in inaccuracies that are not limited to model deficiencies. Attackers can also introduce fake data to compromise machine learning models, resulting in inaccuracies despite the inherent accuracy of the dataset [109] [110] [111].

Large datasets are vulnerable to poisoning attacks in the real world, as discussed by Carlini et al. in [112]. To successfully poison well-known large datasets with hundreds of millions of samples, the authors show the feasibility of two types of poisoning attacks: split-view poisoning and frontrunning poisoning. Another study by Severi et al [60], showcased the vulnerability of different classical software to backdoor poisoning attacks. They showed scenarios involving PDFs, Android apps, and Windows PE files.

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TABLE 3. Example of Backdoor Attack Methods with Poisoning. Refer to Section IV for details on evaluation criteria.

Attack	Scenario	Training Details	Goal	Technique	Evaluation Criteria
BadNets [37]	White-box	Train from scratch	Misclassify	dirty-label	Misclassification rate
Targeted backdoors [58]	Black-box	Train from scratch	Cause	dirty-label	Attack accuracy
Invisible backdoors [88]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate
Reflection backdoors [89]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate
FaceHack [90]	White-box	Train from scratch	Misclassify	dirty-label	Misclassification rate
WarNet [91]	White-box	Train from scratch	Cause	dirty-label	Attack accuracy
Rethink the trigger [92]	White-box	Train from scratch	Misclassify	dirty-label	Misclassification rate
Invisible perturbation [93]	Black-box	Train from scratch	Misclassify	clean-label	Misclassification rate
Bypassing [94]	White-box	Train from scratch	Cause	dirty-label	Attack accuracy
Input-aware [95]	White-box	Train from scratch	Misclassify	dirty-label	Misclassification rate
Dynamic backdoors [69]	White-box	Train from scratch	Misclassify	dirty-label	Misclassification rate
DFST [96]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate
Physical-world backdoors [78]	White-box	Train from scratch	Behave	dirty-label	Accuracy with trojan
Weight perturbations [97]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate
Trojaning attack [98]	White-box	Pre-trained model	Cause	dirty-label	Accuracy with trojan
Latent backdoors [99]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate
Blind backdoors [100]	White-box	Train from scratch	Misclassify	clean-label	Misclassification rate

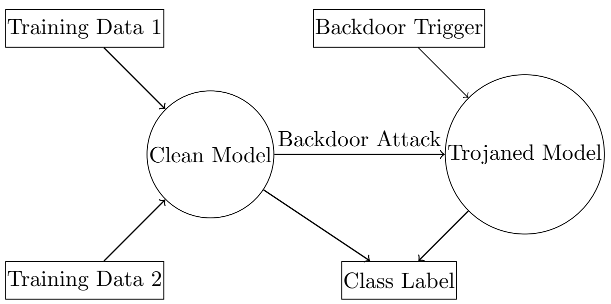

FIGURE 4. Poisoning : Impact of Training Data on the Behavior of deep neural networks .

To protect models from the damage caused by data poisoning, various techniques are available to prevent the penetration of compromised data into and out of the models. Utilizing multiple sources of data, diverse domains, and implementation methods such as data correction, validation, and improved security can strengthen various datasets [113]–[119]. However, attackers can potentially compromise the accuracy of a data-set by manipulating hidden data without detection. Although difficult to perform, such actions can result in incorrect predictions [120].

Although dataset hardening helps prevent data poisoning attacks, attackers can still compromise the accuracy of a data-set by secretly modifying the hidden data. The white box poisoning model has been finalized, but refining the poisoning process of the black box model remains an essential avenue for future research. In addition, viable methods include reducing the number of poisoned samples and, improving the success rate and consistency of poisoning. When evaluating data of unknown origin, it is recommended to use algorithms called " [121]" and " [122]". This will help to identify anomalous samples and pre-process the data to make them more accurate.

Figure 4 depicts a scenario (real-world (e.g., [123]) in which a deep learning model is subjected to a backdoor attack. To execute the attack, the attacker initially trains the model using normal data sets, such as Training Data 1 and Training Data 2. Once the initial training is complete, an attacker introduces a backdoor trigger into the training set. The backdoor trigger is an inconspicuous pattern that is not discernible to humans; however, the model learns to associate it with a specific class label. Despite learning this information, the model behaved as anticipated when evaluated on regular data sets. However, when tested on data containing the backdoor trigger, it incorrectly classified the data as belonging to the target label.

The training process for deep learning systems involves several stages, including data collection, preprocessing, model selection and construction, training, model backup, and deployment. The attackers exploit these steps to launch backdoor attacks. In addition, the dependence of deep learning models on large training datasets and computing resources encourages users to use third-party datasets, platforms or pre-trained models to reduce costs. However, this approach relinquishes control over the training phase, making models more vulnerable to attacks.

Backdoor attacks with poisoning occur mainly during the training (Figure 4) phase of deep-learning systems. During this phase, the neural network adjusts its weights by processing the data and the corresponding labels to generate a model. Attackers modified the training set by introducing triggers and creating poisoned data. They then used this poisoned training set to train the model, ensuring that it remembers these triggers and incorporates backdoors. Consequently, when the poisoned model encounters data containing triggers during the validation phase, it produces the desired label predetermined by the attacker.

III. SUSCEPTIBILITY OF DEEP LEARNING SYSTEMS TO BACKDOOR ATTACKS - ANSWERING RQ1

The susceptibility of deep neural networks to backdoor attacks has been previously investigated. Gu et al. [37] provided a thorough analysis of backdoor attacks in machine learning, including a taxonomy of attacks and their countermeasures, in 2019. The authors discussed several backdoor trigger types, including input-independent, input-dependent, and data-poisoning triggers, as well as methods of network insertion. Techniques to detect and counteract backdoor attacks, including input sanitization, retraining, and fine-tuning, were discussed.

More recently, Ma et al. (2023) [124] proposed a new backdoor attack mechanism called DIHBA, which determines the decision boundaries of the neural network through a dynamic invisible triggering method. The DIHBA has significant resistance to modern backdoor protection systems, a low poisoning rate compared to other backdoor attacks, and a high attack success rate. Wang et al. (2022) [125] used a dataset of chest X-ray images to examine the susceptibility of medical deep learning systems to backdoor attacks. They demonstrated the damaging effects of backdoor attacks on network performance and suggested a strategy based on adversarial training to counteract them.

Using a dataset of voice sounds, another recent study by Zhai et al. (2022) investigated the susceptibility of speaker recognition systems to backdoor attacks. They demonstrated the simplicity of creating backdoor attacks that trick the system [126]. Tian et al. [127] examined the area of machine learning poisoning attacks and provided an outline of the current attack vectors and defense tactics. The authors provided a taxonomy of current defense systems and explored several poisoning approaches, such as label flipping, feature modification, and data injection. Chen et al. [128] focused on backdoor attacks and counterattacks in federated learning, a distributed learning paradigm in which numerous parties cooperate to train a global model. The authors outlined a taxonomy of current backdoor threats and discussed several defense techniques, such as federated learning with differential privacy and model watermarking.

Zhang et al. [87] examined backdoor attacks using deep neural network image classification methods. The authors suggested a defense mechanism that works by fine-tuning the model on a variety of clean datasets, and they showed that it works on numerous benchmark datasets. Wang et al. [81] examined the field of neural trojan attacks. The authors provide an overview of current attack techniques and defense tactics and discuss the significance of creating reliable models to prevent trojan attacks. Guo et al. [129] examined the dangers to pre-trained language models, which have been demonstrated to be open to adversarial and backdoor attacks, among other types of attacks.

To carry out a backdoor poisoning attack more effectively, [130] proposes a method for extracting the crucial areas of the backdoor trigger. The authors explained how to use the activation levels of the neurons to repeatedly update the sounds and how to identify neurons that are more influenced by the backdoor trigger in the classification model. The noise can be employed as a minimum trigger because the discrepancy between the activation values of the important neurons impacted by the backdoor trigger and those by the fully updated noise is minimized.

A clean-label backdoor exploits for anti-spoof rebroadcast detection using a temporal chrominance trigger was proposed in [131]. To identify a specific class of spoofing attacks, particularly video rebroadcast assaults, this study proposes a covert clean-label video backdoor attack against Deep Learning (DL)–based models. Under normal circumstances, the implanted backdoor has no impact on spoofing detection, but it causes misclassification when a certain triggering signal is present. The proposed backdoor depends on a temporal trigger that modifies the average chrominance of a video series.

The research in [132] proposes a data poisoning (DP)–based stealthy and potent backdoor attack against neural networks. In contrast to earlier strikes, this technique uses a sneaky poison and trigger. They can switch a sample's classification in the model from a source class to a target class of the attacker's choice. They achieved this without modifying their labeling by employing a small number of poisoned training samples with barely noticeable perturbations.

Based solely on the knowledge of illustrative cases from the target class, [133], [134] presents an approach to mount clean-label backdoor attacks. When test examples from any class are patched with a backdoor trigger, they can train a model to classify test instances from any class into the target class, with poisoning equal to or less than 0.5% of the target-class data and 0.05% of the training set. Finally, Poisoning MorphNet presents the first backdoor attack technique on point clouds [135]. The suggested backdoor is based on a reconstruction loss that maintains the visual similarity between benign and poisoned point clouds, and a classification loss that requires a modern recognition model of point clouds that tends to misclassify the poisoned sample into a pre-specified target category.

In response to the first research question, we found that data poisoning, backdoor triggers, and Trojan horse attacks are currently the most effective and secure techniques for launching backdoor attacks. In addition, there are other methods for launching backdoor attacks, but they are not always successful. These include model watermarking [136]–[142], which involves placing secret triggers or templates in the model architecture, and fine-tuning attacks . In a fine-tuning attack [143], [143]–[146], a pre-trained model is trained with poisoned data, causing it to adapt to malicious trigger patterns. Finally in transfer learning attacks [147]–[151] , attackers propagate backdoors by first training a model on a source task with backdoors, then modifying it on the target task, possibly by passing hidden triggers to the target model.

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IV. EVALUATION CRITERIA - ANSWERING RQ2

In this section, we present a set of criteria for evaluating backdoor attacks against DNNs based on three aspects: attack accuracy, misclassification rate and accuracy with trojaned samples, the '#' symbol in the formulas represents the 'total number'. The most commonly used statistic to assess the effectiveness of a backdoor attack its accuracy. The calculation is as follows:

$$\textbf{Attack accuracy} = \frac{\#\text{ predictions of target class}}{\#\text{ of total predictions}}.$$

Another statistic was the misclassification rate. It is useful for class specific attacks, in which the backdoor opens only when the trigger is connected to the input of a particular class. Trigger and class are the two conditions required in this case to open the backdoor. The backdoor must not be opened by an input that has the trigger but belongs to a different class than the source class. The formula used to calculate this metric is as follows:

$$\textbf{Misclassification rate} = \frac{\#\text{ wrong predictions}}{\#\text{ of total predictions}}.$$

where the number of incorrect predictions considers poisoned samples that are classified in the target class even though they belong neither to the source nor target class.

As an alternative, it is possible to evaluate accuracy using poisoned samples that do not belong to the source class. This can be evaluated as follows:

$$\textbf{Accuracy with trojan} = \frac{\#\text{ correct predictions}}{\#\text{ of total predictions}}.$$

Although the total number of predictions only considers Trojan samples that do not belong to the source class, the number of correct predictions considers Trojan samples that are correctly sorted into non-source classes. It is crucial to evaluate the drop in clean accuracy, which measures the variation in accuracy provided by the backdoor on clean data because the backdoor must be secret and imperceptible.

V. DEFENCE STRATEGIES AGAINST BACKDOOR ATTACKS - ANSWERING RQ3

A. BACKGROUND

Effective defense against backdoor attacks can be rooted in empirical and validated approaches. Although empirical strategies tend to perform well, there is room for debate regarding their conceptual soundness. Authenticated defenses, however, face theoretical ambiguity in certain circumstances and frequently demonstrate inferior practical performance compared with empirical methods. Current authentication and backdoor defense strategies, as demonstrated by [152] and [153], predominantly depend on random slip technology. However, a variety of approaches are available for empirical backdoor protection strategies, that can be segmented into six primary groups [22]:

These six groups include: (1) preprocessing defenses, which focus on changing trigger patterns; (2) model reconstruction, where erroneous predictions are corrected; (3) trigger reconstruction and synthesis, which concentrates on the elimination of triggers or synthetization of triggers to alter their behaviors, (4) pattern diagnosis-based defense, which employs meta-classifiers to detect a backdoor, (5) sample filtering, to remove contaminated samples; and (6) toxicity inhibition, to inhibit poisoned samples from forming.

Next, we list all available studies that have developed detection methods based on one of these six approaches:

1) **Defense based on preprocessing:** To alter the trigger models present in the attacked samples, these techniques first add a pre-processing module before injecting the samples into DNN networks. Consequently, the secret backdoor can no longer be enabled because the updated triggers no longer match it. Representative works in this domain include [59] , [86] and [92], [86], [154]–[158].
2) **Defense based on model reconstruction:** By directly altering questionable templates, template-based techniques seek to close hidden backdoors in the template. In this manner, the model reconstructs the predictions correctly even if the trigger is present in the attacked samples because the hidden backdoors have already been closed. Representative examples include [59] [159] [160] , [159], [161], [162], [163]–[165]
3) **Defense based on trigger reconstruction or synthesis:** With reconstruction, solutions delete linked triggers to close the backdoor after rebuilding the backdoor trigger in the infected model. Representative examples include [166] , [167] and [168]: For trigger synthesis, defenses first produce a trigger for the backdoor and then eliminate its effects. There are considerable similarities between these defenses and reconstruction-based defenses. Pruning and retraining are the primary strategies used in both types of defenses to close the backdoor, making the elimination process more effective with synthesis-based defenses than with reconstruction-based defenses because the trigger data gathered in synthesis-based defences may be more reliable or sufficient. Representative examples include [167]–[172], [83], [166], [169], [173]–[175], [175], [176], [176]–[181].
4) **Defense based on model diagnosis:** These defenses provide evidence that a suspect model is compromised based on a pre-trained meta-classifier. Because benign models are deployed they can determine whether the model contains any backdoors and refuse to deploy it if it does. Representative examples include [182] [183] [184] , [84], [177], [182]–[193].
5) **Sample filtering:** The goal of these protections is to remove the contaminated samples from the training set. Only benign samples or purified poisoned samples are

used in the learning process after the filtering phase. This stopped the development of backdoors at the source. Representative examples include [194] [195] [196] [197]–[212].

6) **Toxicity inhibition:** Prevents the efficacy of poisoned samples during training by inhibiting their ability to effectively create a backdoor. These works include: [213] and [214].

B. DETECTING BACKDOOR ATTACKS BY APPROXIMATING LOCALIZED CHANGES AND UBIQUITOUS TRANSFORMATION

Backdoor attacks can also be classified according to how they control what occurs in the model. For example, in image models, backdoors that control the pixel space differ from those that control the feature space. The backdoor uses different measurement functions to measure the difference between what is supposed to happen and what actually happens, such as L2 or $L\infty$ norms. There is a detection framework capable of approximating localized changes and ubiquitous transformation to detect backdoor attacks [215].

Metrics are used to quantify the efficiency of an attack. In the case of a patch backdoor attack, for example, the control space refers to the area where the pixels are altered (e.g., by introducing a "trigger" like a patch). In contrast, the backdoor filter employs a distinct control space designated as the style space, which is determined by computing the mean and standard deviation of the input pixels. The metric function employed to assess this attack is L2. To demonstrate the generality of this definition, a discrete Fourier transform was used to map an input image into a frequency space. The L1 metric function was then used to measure performance. In [216], backdoors can target different vulnerable spaces and use different metric functions [215], [217].

Attack instance detection is a method for detecting whether samples contain backdoors. It works well for detecting backdoors inserted into the model but not for detecting naturally created backdoors. Natural backdoors use regular functions that are learned by the model; therefore it is difficult to detect whether they are present in the samples. However, certified robustness aims to obtain correct predictions even if the samples contain backdoors. However, as natural backdoors are common, it is difficult to create effective masks to protect against this type of attack [215]. Two types of backdoors can be used to gain access to the system: a backdoor that identifies a specific word and a backdoor that replaces the word with a synonym. The last type of backdoor is sentence paraphrasing, which uses a style transfer model to transform the original sentences into sentences that facilitate attacks [215], [218].

C. GUARDING AGAINST HOST ATTACKS

To safeguard mission-critical applications that rely on AI models and prioritize privacy and security, it is crucial to strengthen deep neural network models against malicious attacks. Taking a proactive stance provides a robust defense mechanism while clarifying the various methods used in these attacks. Backdoor vulnerabilities are found when an exploitable function is identified, allowing unauthorized access to a model even though the model structure is secure. An attacker exploits this issue by identifying a trigger function within the model, and then discovering another function that maps the input space to a modified regulatory domain.

The closer these two functions are, the greater the risk of exploitation of this vulnerability will be exploited. Notably this vulnerability can be exploited regardless of the resilience of the model structure. It is essential to have knowledge of the various techniques employed by attackers to enhance protection against such attacks. Accordingly, we created two tables that display attack methods and defenses [215] [219], [220].

Table 4 categorises attack methods utilised by adverses based on key characteristics, including the visibility of the method (visible or invisible), the specificity of the attack (sample-specific, clean-label, etc.), and the labelling mechanism used. Noteworthy methods, such as "Composite," "BadNets," and "Blended," are listed, each revealing the unique facets of attacks.

TABLE 4. Attacks techniques used by the adversaries

Attack Method	Visibility	Specificity	Label
Composite [221]	Visible	-	-
BadNets [37]	Visible	-	-
Blended [58]	Invisible	-	-
WaNet [91]	Invisible	-	-
ISSBA [222]	Invisible	Sample specific	-
PhysicalBA [158]	Physical	Sample specific	-
LIRA [223]	Invisible	Sample specific	Optimized
Sleeper [224]	Invisible	Sample specific	Clean label
TUAP [225]	Invisible	Clean label	-
Refool [89]	Sample specific	Clean label	-
UBW-P [226]	Dispersable	controlled	Untargeted
Blind [100]	No optimize	-	-
IAD [95]	Optimized	Sample specific	-
Low [204]	Optimized	Sample specific	-

Table 5 categorizes defense systems based on their ability to respond to attacks. The table includes various defense types, such as pre-processing-based, model repair, poison suppression, sample diagnosis, input perturbation, and model anomalies, each accompanied by specific defense methods. The table also emphasizes the defender's capability in terms of model accessibility (black-box or white-box) and the approach taken (e.g., training from scratch).

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TABLE 5. Classified defense systems by defense capability

Defense Type	Defense Method	Defender Capacity
Pre-processing-based	AutoEncoder [158]	Black-box Model
	ShrinkPad [158]	Black-box Model
Model Repairing	Fine-tuning [159]	White-box Model
	Pruning [159]	White-box Model
	MCR [160]	White-box Model
	NAD [227]	White-box Model
Poison Suppression	ABL [228]	Training from zero
	CutMix [229]	Training from zero
	D-BR,D-ST [198]	Training from zero
	DBD [230]	Training from zero
Sample Diagnosis	SS [194]	Suspicious Dataset
Input Perturbation	NC [166]	Black-box Model
	ABS [231]	Black-box Model
	TABOR [232]	Black-box Model
	STRIP [196]	Black-box Model
	Neo [154]	Black-box Model
Model Anomalies	SentiNet [209]	White-box Model
	SCAn [205]	White-box Model

D. DEFENSE AGAINST THE PARAMETERS OF A BACKDOORED NEURAL NETWORK

This section and Table 5 present some of the methods used to improve the detection of backdoor attacks. As mentioned previously, they fall under six different main defense-type categories. The main defense methods are discussed next.

AutoEncoder: This defense uses autoencoders to reconstruct input data and detect backdoor triggers by comparing the original input with the reconstructed output. Significant disparities between the two indicate the presence of a backdoor trigger.

ShrinkPad: This defense decreases the dimensionality of input data by reducing irrelevant or duplicate elements, making it more difficult for attackers to install backdoor triggers. It is a more effective defense than AutoEncoder since it does not need the learning of an extra model.

FineTuning: This defense fine-tunes the weights of a pre-trained model on a new dataset without backdoor triggers. This overrides backdoor triggers and makes the model more robust against backdoor attacks.

Pruning: This defense eliminates neurons or connections from a neural network that are judged unnecessary. This makes the model simpler and more efficient, but it also makes including backdoor triggers more complex.

MCR (mode connectivity repair): This defense identifies backdoor triggers by examining connectivity patterns in the loss landscape. Mode connectivity is the ability of the loss landscape to connect different local minima.

NAD (neural attention distillation): This defense transfers knowledge from a clean model to a poisoned model using neural attention distillation. This can help remove backdoor triggers from the infected model and enhance its resistance to backdoor attacks.

ABL (anti-backdoor learning): This defense involves training a model with poisoning in the dataset. Backdoor triggers in the poisoned data are designed to activate the backdoor when the model encounters certain inputs. The model can learn to detect and suppress backdoor triggers by training on this data.

SS (Spectral Signature): This defense lies in the rationale that the signals in the learned representation are critical to the classifier's performance. This means that when backdoor attackers aim to mislead the results of classification, they must enhance these signals in representations to make classification deviate from correct outcomes.

SCAn (Statistical Contamination Analyzer): The target contamination attack [233], [234] is a data poisoning attack to poison the samples for specific classes without touching data for other classes. This is hard to be addressed by the general defense for source-specific backdoor attacks. Thus, SCAn relies on distributions to detect any inconsistencies in representation spaces from the backdoored model to figure out the poisoned class in the inputs.

STRIP (STRong Intentional Perturbation): This defense uses the Trojan horse attack strategy to expose the triggers that make input data influential, subsequently weakening classification accuracy. Poisoned data in Trojan horse attacks has lower entropy than clean data. Therefore, STRIP detects and eliminates infected input data efficiently without constraining the size of the triggers during the learning phase.

SentiNet: This defense detects backdoor attempts on DNNs using the following methods: Adversarial object localization [235], [236]. This is used to identify the inputs that have the greatest influence on prediction results, i.e. the most poisoned inputs. SentiNet assesses the quantity of data affected after applying these regions to a clean dataset. The more data that is poisoned, the more likely it is that a region may be attacked.

Neo (model agnostic): This defense detects and mitigates backdoor attacks regardless of the machine learning models that are applied to the classification problem. Neo always assumes backdoored triggers exist at the same but unknown locations on the poisoned data, such as a certain corner of an image. Neo makes use of this information to apply trigger blocks to identify probable backdoored inputs, block backdoor triggers on those inputs, and detect backdoors on the datasets provided.

NC (neural cleanse): This defense detects backdoors by scanning model output labels and reverse-engineering any potential hidden triggers. Once a backdoor has been detected, mitigation efforts are mainly made by filtering out triggered images and patching the affected neurons by unlearning images with the trigger (or using gradient descent [225], [237] to find potential triggers).

CutMix , MaxUp : This defense is based on the assumption that a dramatic increase in data triggers enough perturbation and distortion in the training data to block attacks. This approach is very effective against training-only and backdoor attacks and has the advantage of not degrading the performance of the model. Another trigger reconstruction defense called **TABOR** further improves Neural Cleanse by enhancing the fidelity of backdoor triggers reconstructed via heuristic regularization [174] methods.

VI. BACKDOOR ATTACKS IN OTHER FIELDS - ANSWERING RQ4

A. CHALLENGES ACROSS DOMAINS

Backdoor attacks have been explored across various domains. Below, we list only a few examples.

1) **Acoustic signal processing**: Backdoor attacks have been shown to manipulate audio data, potentially resulting in misclassified speech recognition systems. Examples include [238] [239] [240] [56] [241]–[244] [245]].
2) **Medical sciences**: In the medical field, backdoor attacks aim to compromise medical data, including MRI scans, which can lead to incorrect diagnoses in clinical settings. Examples include [246] [247] [248] and [249].
3) **Cybersecurity**: Attackers can compromise systems by backdooring firewalls, that allow unauthorized network access. Exampels include [250] [60] [158] [251].
4) **Point clouds**: Manipulating 3D data, such as those used in autonomous vehicles, can result in hazardous scenarios, such as deliberate collisions. The studies include [252] [253] , [254] , [255] and [135].
5) **Federated learning**: In federated learning, privacy breaches can occur when malicious actors exploit backdoors in the system, which can extract private information from users. Examples include [256] [257] [258] [259] [260] [261] [262] [263] [264]–[267] [256], [268]–[273].
6) **Transfer learning**: Backdoors can transfer data between models, for example, from a benign dataset-trained model to one with sensitive data. The studies addressing this vulnerability include [274] , [275], [276], [277] , [278] , [147], [99], [279] and [280].
7) **Reinforcement learning**: In reinforcement learning attacks, the rewards given to reinforcement learning agents can be modified, influencing their behavior in favor of an attacker. This work include [281] [282] [283] [284] [285] [286] [287] [288] [289].
8) **Natural language processing**: Text data tampering can result in biased or misleading materials when natural language processing systems are backdoored. Examples include [290] [291] [292] [293] [294] [295] [296] [297]–[299] [62], [63], [144], [173], [298], [300]–[307].

B. CHALLENGES AND OPPORTUNITIES

Through our analysis, we identified several gaps, challenges and open questions in the existing research on backdoor attacks and deep learning vulnerabilities. Some of the main gaps include the limited understanding of attack mechanisms, lack of comprehensive datasets for evaluation and need for more realistic threat models. In addition, the challenges of backdoor transferability between different models and the trade-off between clean model accuracy and backdoor resistance remain open areas for investigation.

The creation and activation of backdoors is the most challenging aspect of backdoor learning. Previous research has not investigated the origins of hidden backdoors or internal processes within infected models when triggers are activated. A deeper understanding of the principles underlying backdoor attacks may contribute to the development of more powerful defense mechanisms.

One efficient method to stop backdoor attacks is adversarial training, which includes adding hostile samples to the training data to strengthen the model's defenses. Another strategy is to create fresh models that can naturally withstand backdoor attacks by incorporating openness and explainability. DNN backdoor attacks are more of a technical problem. These attacks may potentially make use of sophisticated human manipulations and social engineering.

When the data are clean, the backdoor model can still function correctly, which makes it difficult to detect its existence. Backdoor attacks, on the other hand, pose a substantial risk when training is outsourced to a third party. All training resources, including data, models, and training methods, are available to the contracted parties. These third-party employees could be malevolent and smuggle backdoors into the model without user knowledge. Further research is urgently needed to increase the security and dependability of machine learning systems. This is demonstrated by a vast literature review and upcoming developments in backdoor attacks on DNNs.

C. OPEN QUESTIONS

Several open question remain in the domain of backdoor attacks. Below, we list some exemples:

- Understanding the effectiveness of backdoor attacks in the physical world: Physical backdoor attacks involving face recognition have received considerable media attention. However, the effectiveness of such systems varies depending on the surrounding physical conditions. Factors such as lighting or camera angle may have an impact on the success of a backdoor attack. It is difficult to determine the impact of such environmental factors on backdoor attack.

- Combining poisoning and test-time attacks for stronger backdoor attacks: Currently, testing typically includes a backdoor trigger, and no additional input tweaks are required. The efficacy of these attacks can be increased by establishing hostile instances or adding other input alterations. The combination of evasion and backdoor attacks may boost success rates and assist in overcoming problems associated with integrating backdoors.
- Backdoor attacks that persist after end-to-end fine-tuning: When the vast majority of layers in an end-to-end model are subjected to optimization without freezing, the efficacy of transfer learning backdoor approaches is reduced. The incorporation of the backdoor trigger in the model's previous layers, which is frequently changed during the fine-tuning stage in a way that eliminates the backdoor, is what causes this lack of effectiveness. Nevertheless, carrying out backdoor attacks effectively without making significant assumptions about the fine-tuning process in the context of transfer learning is a challenge.
- Backdoor attacks with limited training data: Obtaining clean training data and certain triggers that have been omitted from the model are requirements for a number of current backdoor attack techniques. A significant disadvantage could be the limitation of an attacker's access to the model or the training data. This limitation can be overcome by directly integrating the trigger into the model weights via watermarking strategies. With reduced access to the model or training data, this strategy allows enemies to launch backdoor attacks.
- Architecture-agnostic clean-label attacks: Clean-label backdoor attacks are most effective when the attacker has access to a surrogate model that closely resembles the model of the victim. An attacker can construct backdoored samples to fool the victim model. First, they would need to create a surrogate model to extract the trigger. However, in some circumstances an attacker may not have access to a surrogate model. Improved transferability across multiple model designs is feasible by providing transferable adversarial examples and clean-label attacks that target specific instances. An attacker might exploit this to launch a backdoor attack against a larger number of devices.

D. FUTURE PERSPECTIVES

While the literature on backdoor attacks offers numerous studies covering many domains and different contexts, as a discipline, it is still in its infancy, as several crucial questions have yet to be adequately addressed. In this section, we suggest possible research directions that could lead to future breakthroughs in the field of backdoor attack learning.

First, research on the effects of attack parameters, attack portability across many models and domains, and the creation of stronger defenses are urgently needed. It would also be advantageous to examine the potential use of explainable machine learning techniques to learn more about the behavior of backdoor-triggered models. Moreover, the creation of standardized evaluation frameworks is crucial for backdoor attacks and deep learning vulnerabilities to allow for objective comparison and benchmarking. Such frameworks should incorporate a range of sample datasets, consistent evaluation criteria, and thorough test scenarios that reflect the different types of actual attack scenarios.

In terms of detection, future research should focus on the development of real-time adaptive detection systems, as backdoor attacks become more complex. Investigating techniques to track the activity of models dynamically and identify anomalies or deviations caused by potential backdoor attacks could be an alternative. Graph-based approaches, statistical analyses, and unsupervised learning strategies are potential solutions that work well together. Finding a balance between model accuracy and robustness against backdoor attacks remains quite challenging. Future research should concentrate on strategies for preventing backdoor attacks while minimizing their impact on model accuracy. It would be interesting to investigate how backdoor attacks affect model correctness in this situation.

Adaptive attacks can bypass backdoor defenses, and current defenses are often computationally expensive. Future research could also emphasize designing effective and efficient backdoors, as well as studying black box defenses. Cybercriminals manipulate emotions by exploiting specific triggers [308] [309] [310].

In the context of burgeoning large language models [311], [312], [313], backdoor attacks can have a considerable and unforeseen impact on a model's output, with possibly serious repercussions. For example, once enabled, a revised language translation model may yield erroneous translations, leading to misinterpretations and befuddlement. A modified model, on the other hand, has the potential to inflict severe harm to individuals or organizations, particularly when performing crucial activities such as making financial or medical decisions. Several underlying factors can contribute to backdoor attacks in large language models. Skewed or contaminated training data, for example, may result in the model learning a specific response to a trigger. The susceptibility of the neural architecture to adversarial attacks is another contributing factor. These attacks can be leveraged to include a backdoor trigger in input data. Finally, a malicious actor may exploit the optimization technique utilized in the model training to open a backdoor into the learning process.

The emergence of reverse/proxy/architecture solutions has enabled the secure deployment of deep learning models on a wider scale, especially concerning cloud access security Broker (CASB) / Data loss prevention (DLP). Nonetheless, this progress entails the responsibility of businesses to navigate intricate regulatory frameworks, legal matters, and security protocols that surround deep learning models. Third-party dependencies add an extra risk layer. Backdoor attacks and vulnerabilities in open-source models and toolchains can have widespread consequences. In addition, deep learning models must be resistant to the threat of backdoor attacks, highlighting the need for MLSecOps [314] (MLSecOps refers to the integration of security practices and considerations into the Machine Learning development and deployment process) and rigorous model testing. Misconfigurations of these models can be as damaging as misconfigurations of cloud services. Consequently, ensuring secure configurations is crucial for minimizing the potential risks associated with backdoor attacks in such scenarios.

Finally, securing configurations to mitigate the risks associated with backdoor attacks in cloud services is essential as they can be equally damaging. It is essential to emphasize that the security implications of the adaptable structure of the AI market also apply, particularly for large language models. Whether these models are used as efficient third-party APIs (Application Programming Interface) or form a diverse ecosystem to prevent indirect injection attacks, the involvement of agents, plugins, and multiple system components greatly increases complexity and vulnerability to potential backdoor attacks.

VII. CONCLUSIONS

This survey has provided a background on backdoor learning and has addressed four research questions: What are the latest advances in backdoor attacks; what are the metrics used to gauge success; what are the latest detection methods; and lastly, what challenges, opportunities, open research questions, and future directions exist. The study compiles and organizes examples of commonly utilized backdoor attacks, and proposes a standardized methodology for the evaluation of poisoning, trojan, trigger-based, and triggerless-based backdoor attacks. We also review the current defense methods and analyze the connections between backdoor attacks and different areas of study.

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